

**MAT 426, Class 18 Homework: Spectral Decomposition**  
(Due Thursday, April 16, 2026)

1. Find the spectral decomposition of the matrix  $A$ .

$$(a) \ A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad (b) \ A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix} \quad (c) \ A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad (d) \ A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

2. Find a symmetric  $2 \times 2$  matrix  $A$  with eigenvalues  $\lambda_1 = -1$  and  $\lambda_2 = -2$  with corresponding eigenvectors  $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

3. Find a  $3 \times 3$  matrix with eigenvalues  $\lambda_1 = 1$ ,  $\lambda_2 = -4$ , and  $\lambda_3 = -4$  and corresponding eigenvectors  $\mathbf{v}_1 = \begin{bmatrix} 4 \\ 5 \\ -1 \end{bmatrix}$ ,  $\mathbf{v}_2 = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ , and  $\mathbf{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ .

4. Let  $A$  be an  $n \times n$  matrix, all of whose eigenvalues are real. Adapt the proof of the Spectral Theorem to show that there exist an orthogonal matrix  $Q$  and an upper triangular matrix  $T$  such that  $Q^T A Q = T$ .

5. Determine the quadratic form  $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$  for the given matrix  $A$ .

$$(a) \ A = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix} \quad (b) \ A = \begin{bmatrix} 1 & 3 & -2 \\ 3 & -1 & -1 \\ -2 & -1 & 3 \end{bmatrix} \quad (c) \ A = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 2 & 1 \\ -3 & 1 & 3 \end{bmatrix}$$

6. Find the symmetric matrix  $A$  associated with the given quadratic form.

(a)  $Q(\mathbf{x}) = x_1^2 + 2x_2^2 + 8x_1x_2$

(b)  $Q(\mathbf{x}) = 3x^2 - 3xy - y^2$

(c)  $Q(\mathbf{x}) = x_1^2 - x_2^2 + 3x_3^2 - 6x_1x_2 + 8x_2x_3$

(d)  $Q(\mathbf{x}) = 2x^2 - 3y^2 + z^2 - 4xz + xy + 3yz$